E-Commerce Sales & Customer Analytics: Comprehensive Project Report

1. Project Overview & Infrastructure

This project is an end-to-end Data Analytics solution designed to extract actionable business insights from a transnational retail dataset. It covers the entire data lifecycle: from raw data ingestion and ETL processing in Power Query, to relational data modeling (Star Schema), and finally, dynamic visualization in Power BI.

Data Source & Infrastructure:

- **Source:** Kaggle (Tata Online Retail Dataset)
- **Size:** ~500,000+ rows, 8 columns.
- **Infrastructure:** Raw data (*online_retail.csv*) is version-controlled using **Git LFS** due to its large size (>50MB). The data was imported directly into Power BI via Power Query for the ETL process.

2. Data Profiling & Cleansing (ETL)

Data quality assessment and transformation were performed using Power Query. Data profiling was evaluated using Power Query's *Column Quality*, *Column Distribution*, and *Column Profile* features based on the entire dataset.

Data Dictionary & Transformation Logic: The dataset was rigorously cleaned to ensure accuracy for revenue and customer-level analytics:

Column Name	Data Type	Description & Cleaning Logic
InvoiceNo	Text	Unique transaction identifier, contains no null values. Since the focus is on revenue analysis, transactions containing "C" (cancellations/returns) were explicitly excluded.
StockCode	Text	Product identifier. Verified to contain no errors or null values.
Description	Text	Product name. Less than 1% of records were empty. Since these also lacked a CustomerID and had a UnitPrice = 0 (providing no value for revenue analysis), they were removed.
Quantity	Whole number	Item quantity. Filtered to include only values > 0.
InvoiceDate	Date	Date and time of the transaction. Contains no errors or null values. Extracted temporal dimensions (Year, MonthNumber, MonthName, Quarter) to support time-series revenue analysis (for the dim_date table).
UnitPrice	Currency	Price per unit. Filtered to include only values > 0.
CustomerID	Whole number	Customer identifier. Approximately 25% of records contained null values; these were removed to enable accurate customer-level analytics.

3. Data Modeling & Technical Challenges

The normalized tables were structured into a **Star Schema** to optimize filtering and DAX measure calculations.

- **Fact Table:** *fact_transactions* (contains quantitative data: sales, quantities, dates, and foreign keys).
- **Dimension Tables:** *dim_customers, dim_products, dim_date.*

Technical Challenge Solved: During the creation of the 1-to-Many relationship between *dim_products* and *fact_transactions* via *StockCode*, Power Pivot detected duplicates in the dimension table despite Power Query showing unique values.

- **Root Cause:** Power Query is case-sensitive, whereas Power Pivot is not.
- **Solution:** Applied *TRIM* and *UPPERCASE* functions in Power Query to key columns (*StockCode, CustomerID, InvoiceDate)* to standardize formatting, remove hidden spaces, and successfully build the relationships.

4. Dashboard Construction

The analytical dashboard was built in Power BI, utilizing advanced DAX measures (Total Revenue, Average Order Value, MoM Growth, etc.). It consists of three core pages to answer specific business questions:

- **Page 1: Executive Overview** *Answers: "How is the business performing overall?"* Features high-level KPIs, revenue trends over time, top 10 products, and top-performing geographic regions.
- **Page 2: Customer Analytics** *Answers: "Who are our customers and how should we engage them?"* Highlights Customer Segmentation, Pareto Analysis (identifying the top 20% of customers), and Revenue per Segment.
- **Page 3: Growth & Operations** *Answers: "When do people buy and what drives profitability?"* Focuses on Month-over-Month (MoM) Growth %, seasonal peaks, and revenue concentration risk.

5. DAX Engineering: Pareto Analysis

To identify the top-performing customers driving the majority of the business's revenue (The 80/20 Rule), an advanced Pareto Analysis was implemented. The following DAX measures were engineered to dynamically calculate the cumulative revenue and its percentage contribution:

1. Pareto Cumulative Revenue: This measure calculates the running total of revenue for customers, ranked by their total purchase value. It iterates through the customer table and sums the revenue of all customers whose individual revenue is greater than or equal to the current customer in context.

2. Pareto Percentage (%): This measure determines the cumulative weight of each customer by dividing the cumulative revenue up to that point by the grand total revenue of the entire customer base. This allows for accurate 80/20 visualization.

DAX formulas:

```
Pareto Cumulative Revenue =
VAR CurrentRevenue = [Total Revenue]
RETURN
CALCULATE(
    [Total Revenue],
    FILTER(
        ALL(dim_customers[CustomerID]),
        [Total Revenue] >= CurrentRevenue
    )
)

Pareto % =
DIVIDE(
    [Pareto Cumulative Revenue],
    CALCULATE([Total Revenue], ALL(dim_customers[CustomerID]))
)
```

6. Key Business Insights & Recommendations

Transforming data into actionable business strategies:

1. Revenue Concentration Risk:

- *Insight:* The analysis revealed that a significant portion of total revenue is generated by a small segment of top customers (The 80/20 Rule)
- *Recommendation:* While nurturing these 'Champions' is critical, the business must mitigate risk by running targeted campaigns to convert 'Potential Loyalists' and recent buyers into frequent shoppers to diversify the revenue stream.

2. Seasonality & Inventory Operations:

- *Insight:* Sales exhibit strong seasonality, with massive spikes in transaction volume during specific months.
- *Recommendation:* Optimize inventory levels for top-performing dim_products ahead of these historical peaks and allocate marketing budgets to capitalize on high-traffic periods